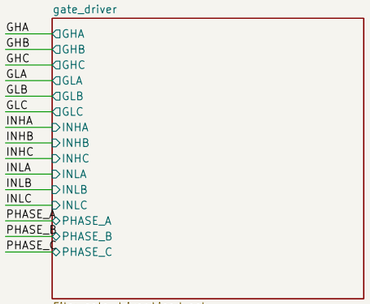
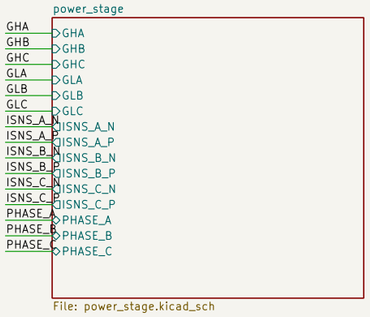
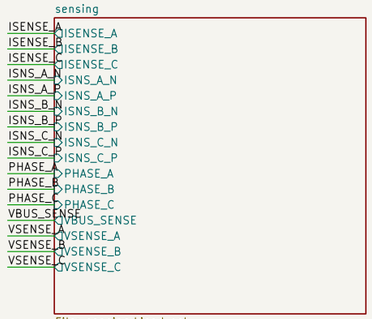
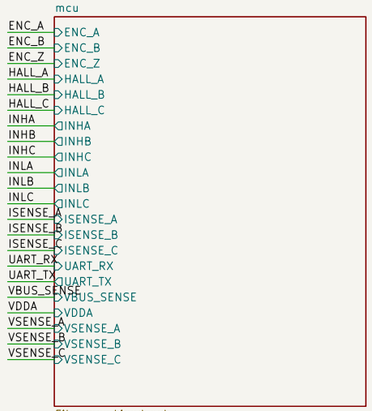
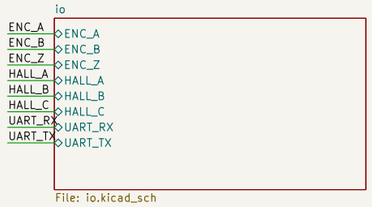












hwde

Sheet: /

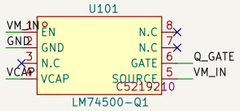
File: PCB-0018-A_bldc-motor-driver.kicad_sch

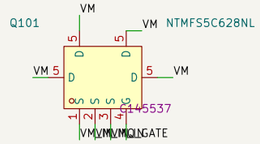
Title: PCB-0018 REV A 3-phase BLDC motor driver: root

Size: A3	Date:	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 1/8

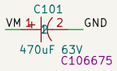


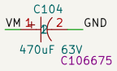


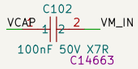


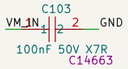


















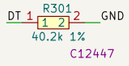
hwde

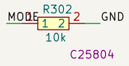
Sheet: /power_in/
File: power_in.kicad_sch

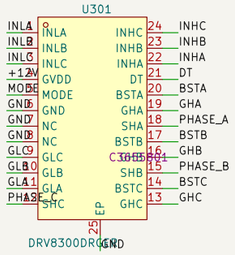
Title: PCB-0018 REV A power_in: reverse-polarity protected VM entry

Size: A3	Date: 2026-09-27	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 2/8

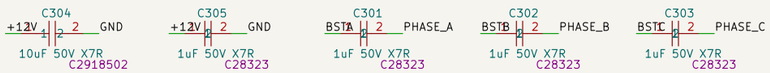
U201
LM5164
GND
VM
RON
EN/UVLOPGOOD
EP
SW
BST
BUCK_SW
BST5V
FB5V
C47906
R201
31.6k 1%
C25967
R202
33.2k 1%
C23003
R203
10k
C25804
22uH shielded
L201
C2849539
C201
2.2nF 50V X7R
C1604
C202
22uF 25V X5R
C45783
R206
100k
C25803
C210
2.2nF 50V X7R
C1604
C211
100nF 50V X7R
C14663
C203
100nF 50V X7R
C14663
C204
10uF 50V X7R
C2918502
22uH shielded
L202
C135264
U202
TPS61040
C7722
D201
MBR0520L
C5563735
C206
1uF 50V X7R
C28323
C213
100nF 50V X7R
C14663
C205
10uF 25V X5R
C15850
R204
86.6k 1%
C23121
C212
1nF 50V X7R
C1588
R205
10k
C25804
U203
TLV75733
C495517
C208
1uF 50V X7R
C28323
C207
1uF 50V X7R
C28323
600ohm@100MHz ferrite
FB201
VDDA
VM
PWR_FLAG
GND
hwde
Sheet: /regulators/
File: regulators.kicad_sch
Title: PCB-0018 REV A regulators: VM->5V buck, 5V->12V boost, 5V->3V3
Size: A3
Date: 2026-09-27
Rev: 1
KiCad E.D.A. 10.0.6
Id: 3/8













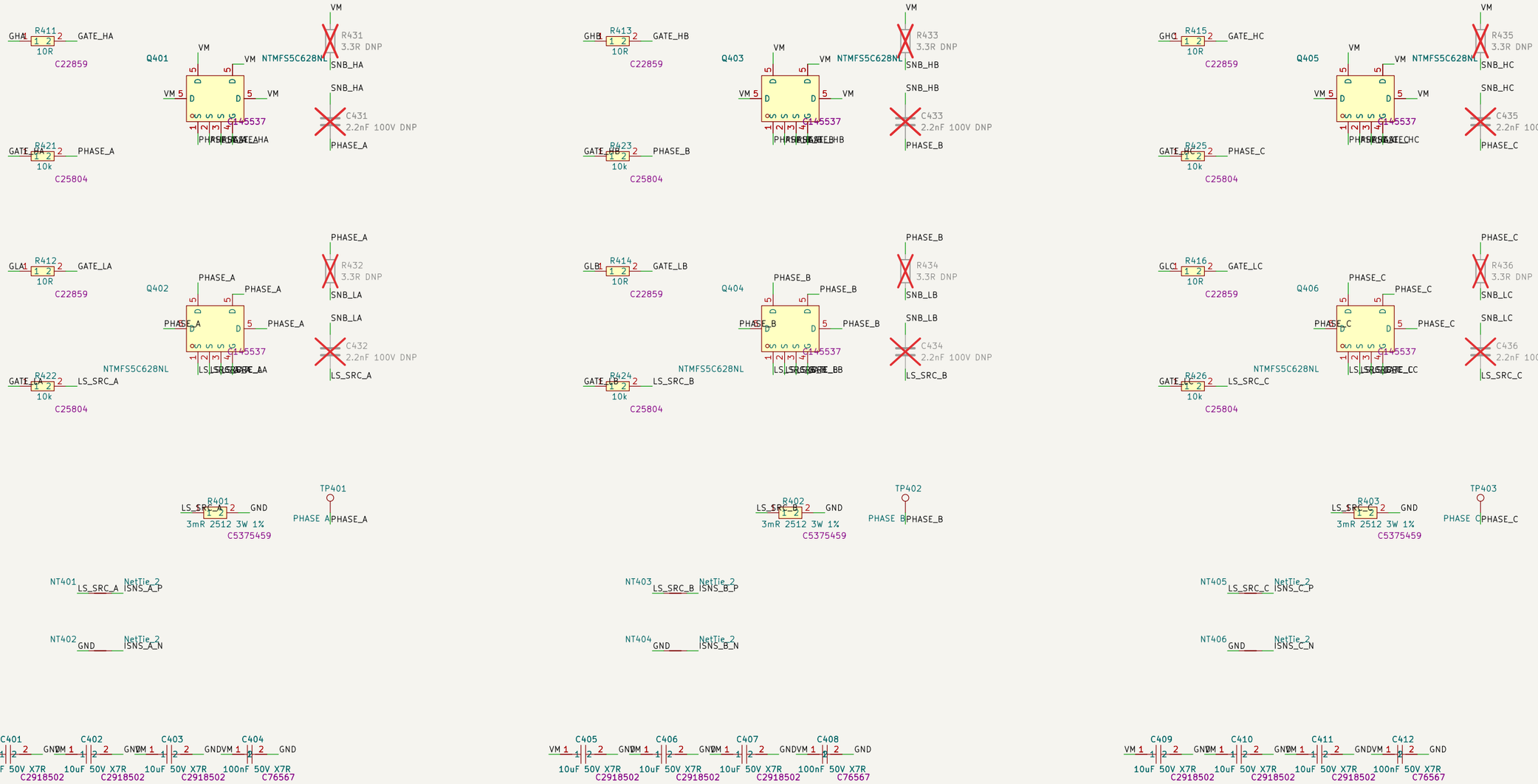


hwde

Sheet: /gate_driver/
File: gate_driver.kicad_sch

Title: PCB-0018 REV A gate driver: DRV8300D, GVDD 12 V, 6-PWM, 200 ns

Size: A3	Date: 2026-09-27	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 4/8



hwde		
Sheet: /power_stage/		
File: power_stage.kicad_sch		
Title: PCB-0018 REV A power stage: 3x NTMFS5C628NL half-bridge, 3 mOhm		
Size: A3	Date: 2026-09-27	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 5/8

hwde

Sheet: /sensing/

File: sensing.kicad_sch

Title: PCB-0018-A bldc-motor-driver: sensing (3x INA240A1 shunt CSA + p

Size: A3	Date: 2026-09-27	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 6/8

VDDA
VBUS_SENSE
ISENSE_A
ISENSE_B
ISENSE_C
VSENSE_A
VSENSE_B
VSENSE_C
HALL_A
HALL_B
HALL_C
ENC_A
ENC_B
ENC_Z
UART_RX

INHA
INHB
INHC
INLA
INLB
INLC
UART_TX

U601 STM32G431CBT6 C529355

+3V3 VDD VDD VDD VDDA

NRST PG10-NRST
PF0-OSC_IN
PF1-OSC_OUT
VBAT
VREF+
USER_SW PC13
PC14-OSC32_IN
PC15-OSC32_OUT
VBUS_SENSE PA0
ISENSE_A PA1
VSENSE_A PA2
VSENSE_B PA3
VSENSE_C PA4
PA5
PA6
ISENSE_B PA7
INHA PA8
INHB PA9
INHC PA10
PA11
PA12
SWDIO PA13
SWCLK PA14
HALL_A PA15

PB0 ISENSE_C
PB1
PB2
PB3 HALL_B
PB4 ENC_A
PB5 ENC_B
PB6 UART_TX
PB7 UART_RX
PB8-BOOT0 BOOT0
PB9 ENC_Z
PB10 HALL_C
PB11 LED_STATUS
PB12 LED_FAULT
PB13 INLA
PB14 INLB
PB15 INLC

VSS VSS VSS VSSA
GND GND GND GND
GND GND

C609 NRST GND 100nF 50V X7R C14663

SW601 NRST GND Reset switch SMD C720477

R601 BOOT0 GND 10k 1% C25804

SW602 USER_SW GND User switch SMD C720477

C610 USER_SW GND 100nF 50V X7R C14663

R602 +3V3 LED_D601_A 1k 1% C21190
D601 GND2 LED_D601_A LED green (power) C72041

R603 LED_STATUS LED_D602_A 1k 1% C21190
D602 GND2 LED_D602_A LED yellow (status) C2286

R604 LED_FAULT LED_D603_A 1k 1% C21190
D603 GND2 LED_D603_A LED red (fault) C2286

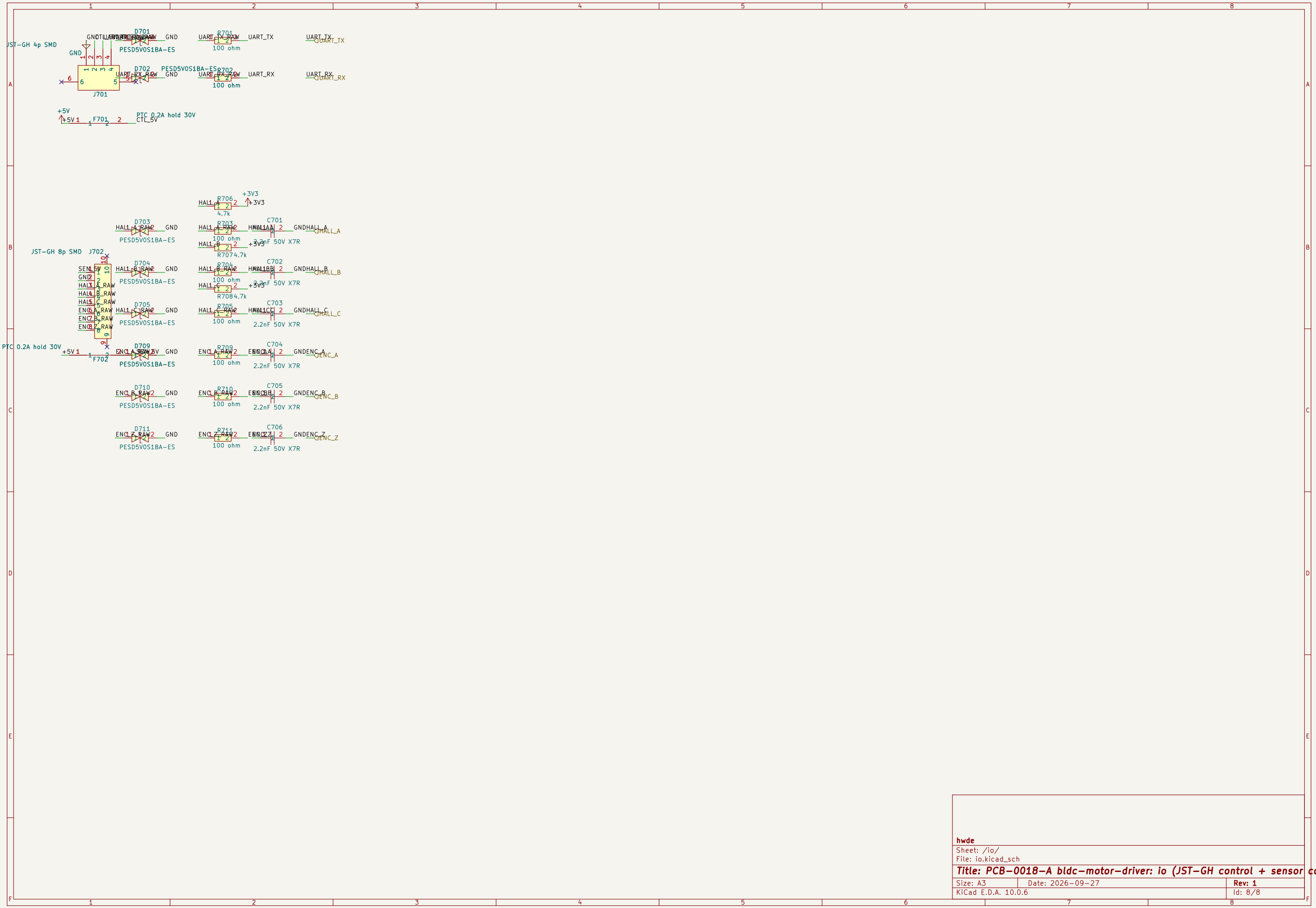
J601 SWD 2x5 1.27mm C2935458
+3V3 GND GND GND
SWDIO SWCLK NRST

C601 +3V3 GND 100nF 50V X7R C14663
C602 +3V3 GND 100nF 50V X7R C14663
C603 +3V3 GND 100nF 50V X7R C14663
C604 +3V3 GND 4.7uF 25V X5R C1779
C605 VDDA GND 1uF 50V X7R C28323

VDDA
C606 10nF 50V X7R GND C57112

C607 VDDA GND 1uF 50V X7R C28323
C608 VDDA GND 100nF 50V X7R C14663

hwde
Sheet: /mcu/
File: mcu.kicad_sch
Title: PCB-0018-A bldc-motor-driver: mcu (STM32G431CBT6)
Size: A3 | Date: 2026-09-27 | Rev: 1
KiCad E.D.A. 10.0.6 | Id: 7/8



hwde		
Sheet: /io/		
File: io.kicad_sch		
Title: PCB-0018-A bldc-motor-driver: io (JST-GH control + sensor connect		
Size: A3	Date: 2026-09-27	**Rev: 1**
KiCad E.D.A. 10.0.6		Id: 8/8